

R PROGRAMS

UNIT-1

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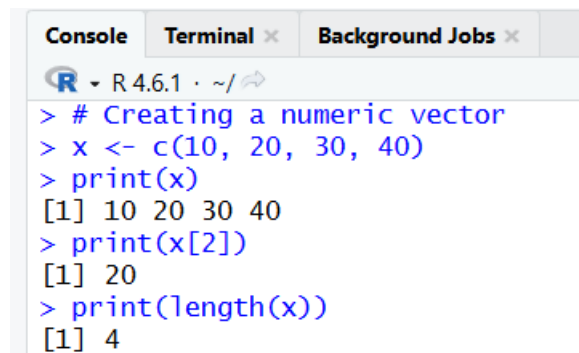
1. Vectors in R

PROGRAM

```
# Creating a numeric vector  
x <- c(10, 20, 30, 40)
```

```
print(x)  
print(x[2])    # Access second element  
print(length(x)) # Number of elements
```

OUTPUT



```
R - R 4.6.1 - ~/   
> # Creating a numeric vector  
> x <- c(10, 20, 30, 40)  
> print(x)  
[1] 10 20 30 40  
> print(x[2])  
[1] 20  
> print(length(x))  
[1] 4
```

2. Atomic Vectors

PROGRAM

```
# Atomic vector example  
a <- c(1, 2, 3)  
print(a)  
print(typeof(a))  
print(length(a))
```

OUTPUT

```

> # Atomic vector example
> a <- c(1, 2, 3)
> print(a)
[1] 1 2 3
> print(typeof(a))
[1] "double"
> print(length(a))
[1] 3

```

3. Character Vectors

PROGRAM

```

# Character vector
names <- c("Anu", "Ravi", "Meena")
print(names)
print(toupper(names))
print(tolower(names))
print(paste(names, "Student"))

```

OUTPUT

```

> # Character vector
> names <- c("Anu", "Ravi", "Meena")
>
> print(names)
[1] "Anu" "Ravi" "Meena"
> print(toupper(names))
[1] "ANU" "RAVI" "MEENA"
> print(tolower(names))
[1] "anu" "ravi" "meena"
> print(paste(names, "Student"))
[1] "Anu Student" "Ravi Student" "Meena Student"

```

4. Operations on Logical Vectors

PROGRAM

```

# Logical operations
x <- c(TRUE, FALSE, TRUE)
y <- c(FALSE, TRUE, TRUE)
print(x & y)
print(x | y)
print(!x)

```

OUTPUT

```

> # Logical operations
> x <- c(TRUE, FALSE, TRUE)
> y <- c(FALSE, TRUE, TRUE)
> print(x & y)
[1] FALSE FALSE TRUE
> print(x | y)
[1] TRUE TRUE TRUE
> print(!x)
[1] FALSE TRUE FALSE

```

5. Calculating with R

PROGRAM

```
# Arithmetic calculations
```

```
a <- 10
```

```
b <- 5
```

```
print(a + b)
```

```
print(a - b)
```

```
print(a * b)
```

```
print(a / b)
```

```
print(a ^ b)
```

```
print(sqrt(a))
```

OUTPUT

```

[1] TRUE TRUE FALSE
> # Arithmetic calculations
> a <- 10
> b <- 5
> print(a + b)
[1] 15
> print(a - b)
[1] 5
> print(a * b)
[1] 50
> print(a / b)
[1] 2
> print(a ^ b)
[1] 1e+05
> print(sqrt(a))
[1] 3.162278

```

6. Creating Matrices and Operations

PROGRAM

```
# Matrix creation
```

```
m <- matrix(1:6, nrow = 2, ncol = 3)
```

```
print(m)
```

```
print(t(m))
print(m + m)
print(m * m)
```

OUTPUT

```
> # Matrix creation
> m <- matrix(1:6, nrow = 2, ncol = 3)
> print(m)
      [,1] [,2] [,3]
[1,]    1    3    5
[2,]    2    4    6
> print(t(m)) # Transpose
      [,1] [,2]
[1,]    1    2
[2,]    3    4
[3,]    5    6
> print(m + m) # Matrix addition
      [,1] [,2] [,3]
[1,]    2    6   10
[2,]    4    8   12
> print(m * m) # Element-wise multiplication
      [,1] [,2] [,3]
[1,]    1    9   25
[2,]    4   16   36
```

7. Creating Arrays and Operations

PROGRAM

```
# Array creation
a <- array(1:12, dim = c(2, 3, 2))

print(a)

print(a + 5)

print(a * 2)

print(dim(a))
```

OUTPUT

```

> # Array creation
> a <- array(1:12, dim = c(2, 3, 2))
> print(a)
, , 1
      [,1] [,2] [,3]
[1,]    1    3    5
[2,]    2    4    6

, , 2
      [,1] [,2] [,3]
[1,]    7    9   11
[2,]    8   10   12

> print(a + 5)
, , 1
      [,1] [,2] [,3]
[1,]    6    8   10
[2,]    7    9   11

, , 2
      [,1] [,2] [,3]
[1,]   12   14   16
[2,]   13   15   17

> print(a * 2)
, , 1
      [,1] [,2] [,3]
[1,]    2    6   10
[2,]    4    8   12

, , 2
      [,1] [,2] [,3]
[1,]   14   18   22
[2,]   16   20   24

```

8. S3 Class in R

PROGRAM

```

# S3 Class
obj <- list(name = "Anu", marks = 80)
class(obj) <- "student"
print(obj)
print(class(obj))

```

OUTPUT

```

> # S3 Class
> obj <- list(name = "Anu", marks = 80)
> class(obj) <- "student"
> print(obj)
$name
[1] "Anu"

$marks
[1] 80

attr(,"class")
[1] "student"
> print(class(obj))
[1] "student"

```

9. S4 Class in R

PROGRAM

```

# S4 Class
setClass("Student",
        slots = list(name = "character", marks = "numeric"))
s1 <- new("Student", name = "Anu", marks = 80)
print(s1)

```

OUTPUT

```

> # S4 Class
> setClass("Student",
+         slots = list(name = "character", marks = "numeric"))
>
> s1 <- new("Student", name = "Anu", marks = 80)
>
> print(s1)
An object of class "Student"
Slot "name":
[1] "Anu"

Slot "marks":
[1] 80

```

10. Reference Class in R

PROGRAM

```

# Reference Class
StudentRC <- setRefClass(
    "StudentRC",
    fields = list(name = "character")
)

```

```
)  
s1 <- StudentRC$new(name = "Anu")  
print(s1$name)  
s1$name <- "Ravi"  
print(s1$name)
```

OUTPUT

```
> # Reference Class  
> StudentRC <- setRefClass(  
+   "StudentRC",  
+   fields = list(name = "character")  
+ )  
> s1 <- StudentRC$new(name = "Anu")  
> print(s1$name)  
[1] "Anu"  
> s1$name <- "Ravi"  
> print(s1$name)  
[1] "Ravi"  
> |
```